

Topics in Geometry and Topology: Assignment 5

钟皓宇(12533556)

May 17, 2026

Problem 1

Show that any two-dimensional complete toric variety is projective. Show that any ample divisor on such a variety is very ample.

Solution: Let X_Σ be an two-dimensional complete toric variety. We show that Σ admits a strictly convex integral support function ψ . Order the rays of X_Σ clockwise as

$$\rho_i = \mathbb{R}_{\geq 0} \cdot v_i \quad i = 0, \dots, r$$

where v_i denotes the primitive generator of ρ_i . We let

$$\sigma_i = \langle v_i, v_{i+1} \rangle \quad i = 0, \dots, r-1$$

and let $\sigma_r = \langle v_0, v_r \rangle$ be maximal cones of Σ . For each i , choose a primitive generator

$$u_i \in \rho_i^\perp \cap \sigma_i^\vee \cap M.$$

I claim that there exists integers $c_i > 0$ such that

$$\sum_{i=0}^r c_i u_i = 0.$$

If not, according to the separating hyperplane theorem for convex sets, there exists a line passing through origin such that all u_i is contained in a closed half-plane. It is impossible, since it is equivalent to say that all ρ_i lie in a closed half-plane contradicting to the completeness. Next, we choose an arbitrary $m_0 \in \sigma_0^\vee \cap M$, and define

$$m_i = m_{i-1} + c_i u_i.$$

Hence, we consider a support function ψ such that $\psi|_{\sigma_i} = \langle m_i, - \rangle$. Clearly it is continuous since

$$\langle m_i - m_{i-1}, \rho_i \rangle = 0$$

implies that $\psi|_{\sigma_i}$ and $\psi|_{\sigma_{i-1}}$ agree on ρ_i . Moreover, since $c_i > 0$ and $u_i \in \sigma_i^\vee$, we obtain that

$$\langle m_i - m_{i-1}, n \rangle > 0 \text{ for } n \in \sigma_i \quad \langle m_i - m_{i-1}, n \rangle < 0 \text{ for } n \in \sigma_{i-1}$$

which ψ is strictly convex. Therefore X_Σ is projective.

Now let D be an ample divisor on X_Σ . We may replace D by a linearly equivalent torus-invariant Cartier divisor. Write

$$D = \sum_{\rho \in \Sigma(1)} a_\rho D_\rho$$

and define

$$P_D = \{m \in M_\mathbb{R} \mid \langle m, v_\rho \rangle \geq -a_\rho \text{ for all } \rho \in \Sigma(1)\}.$$

Since D is ample, P_D is a lattice polygon and its normal fan is Σ .

By the toric criterion for very ampleness, it is enough to show that for every maximal cone $\sigma \in \Sigma$, the semigroup

$$\sigma^\vee \cap M$$

is generated by

$$(P_D \cap M) - m_\sigma,$$

where $m_\sigma \in P_D \cap M$ is the vertex corresponding to σ .

Fix such a cone σ . Translate P_D by $-m_\sigma$, and put

$$Q = P_D - m_\sigma.$$

Then 0 is a vertex of Q , and the tangent cone of Q at 0 is $\sigma^\vee$. Let $e_1, e_2 \in M$ be the primitive lattice directions of the two edges of Q starting at 0. Then

$$\sigma^\vee = \mathbb{R}_{\geq 0}e_1 + \mathbb{R}_{\geq 0}e_2,$$

and $e_1, e_2 \in Q \cap M$.

We claim that $\sigma^\vee \cap M$ is generated by $Q \cap M$. By convexity, the closed triangle with vertices $0, e_1, e_2$ lies in Q . List the primitive lattice vectors in this triangle in angular order:

$$w_0 = e_1, \quad w_1, \dots, \quad w_s = e_2.$$

For each j , the pair of vectors w_j, w_{j+1} form a $\mathbb{Z}$ -basis of M . Indeed, if not, the parallelogram spanned by w_j, w_{j+1} contains a nonzero lattice point not in the lattice generated by them. Then either this point, or its translate by $w_j + w_{j+1}$, gives a lattice direction strictly between w_j and w_{j+1} , contradicting the choice of consecutive w_j, w_{j+1} .

Hence every lattice point in

$$\mathbb{R}_{\geq 0}w_j + \mathbb{R}_{\geq 0}w_{j+1}$$

is a non-negative integral combination of w_j and w_{j+1} . Since these cones cover $\sigma^\vee$, every element of $\sigma^\vee \cap M$ is generated by the w_j 's. But each w_j lies in the triangle with vertices $0, e_1, e_2$, hence in $Q \cap M$. Thus $\sigma^\vee \cap M$ is generated by $Q \cap M = (P_D \cap M) - m_\sigma$.

Therefore the toric criterion for very ampleness is satisfied for every maximal cone σ , and hence D is very ample. ■

Problem 2

If $X(\Delta)$ is complete and nonsingular, show that a T -divisor is ample if and only if it is very ample.

Solution: Since very ample clearly implies ample, we just prove that ample implies very ample.

Let $D = \sum_{\rho \in \Delta(1)} a_\rho D_\rho$ be a ample T -divisor on $X(\Delta)$. Since $X(\Delta)$ is nonsingular, every T -divisor is Cartier. For each maximal cone $\sigma \in \Delta$, let $m_\sigma \in M$ be the Cartier data of D , so that

$$\langle m_\sigma, v_\rho \rangle = -a_\rho \quad \text{for all } \rho \subset \sigma.$$

Let

$$P_D = \{m \in M_{\mathbb{R}} \mid \langle m, v_\rho \rangle \geq -a_\rho \text{ for all } \rho \in \Delta(1)\}.$$

Since D is ample, m_σ is the vertex of P_D corresponding to σ .

By the toric criterion for very ampleness, it is enough to show that, for every maximal cone σ ,

$$\sigma^\vee \cap M$$

is generated by

$$(P_D \cap M) - m_\sigma.$$

Fix a maximal cone

$$\sigma = \langle v_1, \dots, v_n \rangle.$$

Since $X(\Delta)$ is nonsingular, $v_1, \dots, v_n$ form a $\mathbb{Z}$ -basis of N . Let

$$u_1, \dots, u_n$$

be the dual basis of M , so

$$\langle u_i, v_j \rangle = \delta_{ij}.$$

Then

$$\sigma^\vee \cap M = \mathbb{Z}_{\geq 0} u_1 + \dots + \mathbb{Z}_{\geq 0} u_n.$$

Thus it suffices to prove that

$$u_i \in (P_D \cap M) - m_\sigma$$

for every i . Fix i . Let

$$\tau_i = \langle v_1, \dots, \widehat{v_i}, \dots, v_n \rangle$$

be the codimension-one face of σ obtained by deleting v_i . Since $X(\Delta)$ is complete, there is a maximal cone $\sigma'_i \neq \sigma$ such that

$$\sigma \cap \sigma'_i = \tau_i.$$

Let $m_{\sigma'_i}$ be the Cartier datum of D on σ'_i .

Since $m_{\sigma'_i}$ and m_σ agree on τ_i , we have

$$\langle m_{\sigma'_i} - m_\sigma, v_j \rangle = 0 \quad \text{for all } j \neq i.$$

Hence

$$m_{\sigma'_i} - m_\sigma = c_i u_i$$

for some integer c_i . Since D is ample, the support function is strictly convex, so $c_i > 0$.

Both m_σ and $m_{\sigma'_i}$ are vertices of P_D . Since P_D is convex, the line segment between them lies in P_D . But

$$m_\sigma + u_i = \left(1 - \frac{1}{c_i}\right) m_\sigma + \frac{1}{c_i} m_{\sigma'_i},$$

so

$$m_\sigma + u_i \in P_D \cap M.$$

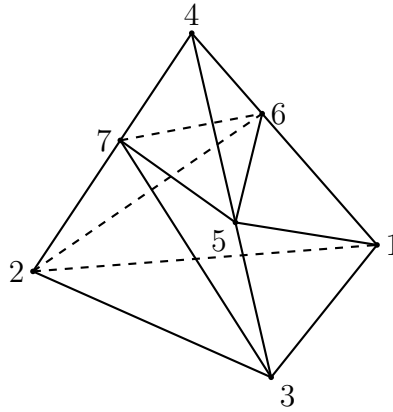
Therefore

$$u_i = (m_\sigma + u_i) - m_\sigma \in (P_D \cap M) - m_\sigma.$$

■

Problem 3

Form the three-dimensional fan Δ whose edges in $\mathbb{Z}^3$ pass through $v_1 = -e_1$, $v_2 = -e_2$, $v_3 = -e_3$, $v_4 = e_1 + e_2 + e_3$, $v_5 = v_3 + v_4$, $v_6 = v_1 + v_4$, and $v_7 = v_2 + v_4$ and with cones through the faces of the triangulated tetrahedron shown below:



Show that the toric variety associated to the fan above is complete and nonsingular, and determine whether it is projective or not.

Solution: It is complete and nonsingular but it is not projective. We use notation Δ_{ijk} to denote the triangulated tetrahedron generated by vertex v_i, v_j, v_k and origin. Let Δ_{0123} denote the triangulated tetrahedron generated by vertices v_1, v_2, v_3, v_4 .

To see the completeness, The decomposition

$$\Delta_{0123} = \Delta_{123} \cup \Delta_{135} \cup \Delta_{156} \cup \Delta_{456} \cup \Delta_{237} \cup \Delta_{357} \cup \Delta_{457} \cup \Delta_{126} \cup \Delta_{267} \cup \Delta_{467}.$$

implies the completeness. Since it gives a subdivision of Δ_{0123} . Let Y be the toric variety corresponding to the fan generated by $\Delta_{123}, \Delta_{134}, \Delta_{234}, \Delta_{124}$. Clearly, Y is complete and $X(\Delta)$ as the blow-up of Y maintains the completeness.

Let determinant of v_i, v_j, v_k denoted by d_{ijk} . Then

$$d_{123} = d_{267} = d_{135} = d_{156} = d_{457} = -1 \quad d_{126} = d_{467} = d_{456} = d_{237} = d_{357} = 1$$

implies the smoothness of $X(\Delta)$.

Finally, we show that $X(\Delta)$ is not projective. Suppose, for contradiction, that $X(\Delta)$ is projective. Then Δ admits a strictly convex support function ψ . Put

$$a_i = \psi(v_i).$$

Using the linear relations

$$v_1 + v_7 = v_2 + v_6, \quad v_3 + v_6 = v_1 + v_5, \quad v_2 + v_5 = v_3 + v_7,$$

and the strict convexity of ψ , we get

$$a_1 + a_7 < a_2 + a_6, \quad a_3 + a_6 < a_1 + a_5, \quad a_2 + a_5 < a_3 + a_7.$$

Adding these three inequalities gives

$$(a_1 + a_7) + (a_3 + a_6) + (a_2 + a_5) < (a_2 + a_6) + (a_1 + a_5) + (a_3 + a_7),$$

which is impossible, since the two sides are equal. Hence no strictly convex support function exists on Δ . Therefore $X(\Delta)$ is not projective. $\blacksquare$

Problem 4

For the toric variety $X = \mathbb{P}^n$ with its divisors $D_0, \dots, D_n$, verify directly that:

1. $D = \sum a_i D_i$ is generated by its sections $\iff \Psi_D$ is convex $\iff a_0 + \dots + a_n \geq 0$,
and
2. D is ample $\iff \Psi_D$ is strictly convex $\iff a_0 + \dots + a_n > 0$.

Solution: (1) Let $X = \mathbb{P}^n$ with its standard fan. The rays are generated by $v_0 = -e_1 - \dots - e_n$ and $v_i = e_i$ for $1 \leq i \leq n$. The maximal cone σ_i is generated by all v_j with $j \neq i$. Let $D = \sum_{i=0}^n a_i D_i$, and put $s = a_0 + \dots + a_n$.

For the Cartier datum $m_i = m_{\sigma_i}$, we have $\langle m_i, v_j \rangle = -a_j$ for $j \neq i$. Hence $m_0 = (-a_1, \dots, -a_n)$, and for $i \geq 1$,

$$m_i = (-a_1, \dots, -a_{i-1}, s - a_i, -a_{i+1}, \dots, -a_n).$$

Thus $m_i - m_0 = se_i^*$.

Now

$$P_D = \{m \in M_{\mathbb{R}} \mid \langle m, v_i \rangle \geq -a_i \text{ for all } i\}.$$

The divisor D is generated by global sections iff $m_i \in P_D$ for every maximal cone σ_i . For m_0 , the only inequality to check is the one for v_0 , and it gives $s \geq 0$. For m_i with $i \geq 1$, the only inequality to check is the one for v_i , and it also gives $s \geq 0$. Therefore

$$D \text{ is generated by sections} \iff \Psi_D \text{ is convex} \iff s \geq 0.$$

(2) For strict convexity, the Cartier data on adjacent maximal cones must be distinct and bend strictly. Since $m_i - m_0 = se_i^*$, this happens exactly when $s > 0$. Hence

$$\Psi_D \text{ is strictly convex} \iff s > 0.$$

Finally, since $D_i \sim H$ for all i , we have $D \sim sH$. Thus $\mathcal{O}(D) \simeq \mathcal{O}_{\mathbb{P}^n}(s)$, which is ample iff $s > 0$. Hence

$$D \text{ is ample} \iff \Psi_D \text{ is strictly convex} \iff a_0 + \dots + a_n > 0.$$

$\blacksquare$

Problem 5

Let

$$A = \{P \subseteq M_{\mathbb{R}} \mid P \text{ is a full dimensional rational polytope}\},$$

$$B = \{(X(\Delta), D) \mid D \text{ is a } T_N\text{-invariant ample divisor in } X(\Delta)\},$$

where Δ is a complete fan, i.e. $|\Delta| = N_{\mathbb{R}}$. Show that the maps $P \mapsto (X_P, D_P)$ and $(X(\Delta), D) \mapsto P_D$ define bijections between the sets A and B which are inverses of each other.

Solution: Let $P \subset M_{\mathbb{R}}$ be a full dimensional rational polytope. Let Δ_P be its normal fan. Since P is full dimensional, Δ_P is complete. For every ray $\rho \in \Delta_P(1)$, let $v_\rho \in N$ be its primitive generator, and define

$$a_\rho = -\min_{m \in P} \langle m, v_\rho \rangle.$$

Set

$$D_P = \sum_{\rho \in \Delta_P(1)} a_\rho D_\rho.$$

Then

$$P = \{m \in M_{\mathbb{R}} \mid \langle m, v_\rho \rangle \geq -a_\rho \text{ for all } \rho \in \Delta_P(1)\}.$$

Indeed, these are exactly the facet inequalities of P . Hence

$$P_{D_P} = P.$$

Moreover, the support function

$$\Psi_P(v) = \min_{m \in P} \langle m, v \rangle$$

is strictly convex with respect to the normal fan Δ_P . Therefore D_P is ample on $X(\Delta_P)$. Thus the map

$$P \mapsto (X_P, D_P)$$

is well-defined.

Conversely, let $(X(\Delta), D) \in B$, where

$$D = \sum_{\rho \in \Delta(1)} a_\rho D_\rho$$

is an invariant ample divisor. Define

$$P_D = \{m \in M_{\mathbb{R}} \mid \langle m, v_\rho \rangle \geq -a_\rho \text{ for all } \rho \in \Delta(1)\}.$$

Since D is ample, its support function Ψ_D is strictly convex with respect to Δ . Hence P_D is a full dimensional rational polytope and its normal fan is exactly Δ .

We now check that the two constructions are inverse. Starting from P , the definition of D_P gives

$$P_{D_P} = P.$$

Starting from $(X(\Delta), D)$, since D is ample, the normal fan of P_D is Δ . Moreover, for every ray $\rho \in \Delta(1)$,

$$\min_{m \in P_D} \langle m, v_\rho \rangle = -a_\rho,$$

because the inequality $\langle m, v_\rho \rangle \geq -a_\rho$ defines the corresponding facet of P_D . Therefore the divisor associated to P_D is

$$D_{P_D} = \sum_{\rho \in \Delta(1)} a_\rho D_\rho = D.$$

Thus

$$P \mapsto (X_P, D_P), \quad (X(\Delta), D) \mapsto P_D$$

are inverse bijections. ■